

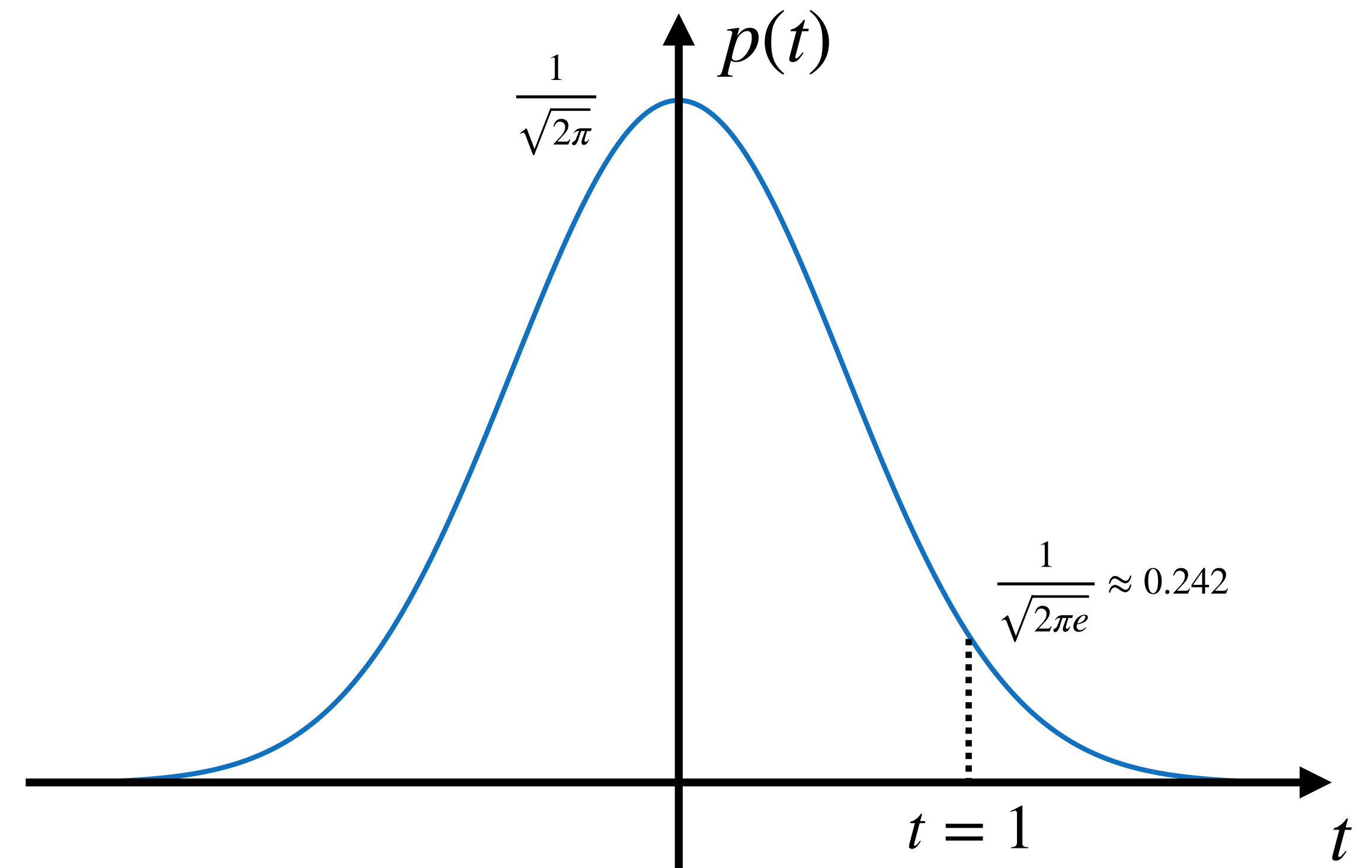
EE121 Discussion 7

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What's the Probability?

$$X \sim \mathcal{N}(0,1), \quad \begin{cases} P(X > x) = Q(x) = \int_x^\infty p(t) dt \\ P(X \leq x) = \Phi(x) = \int_{-\infty}^x p(t) dt \end{cases}$$

What's $p(t)$?



I'm given $Y \sim \mathcal{N}(\mu, \sigma^2)$, can
I use $\Phi(x)$ and $Q(x)$?

How are $Q(x)$ and $\Phi(x)$ related?

What's the Probability?

$$Y \sim \mathcal{N}(8,4), P(Y > 6) =$$

$$Y \sim \mathcal{N}(-3,1), P(Y \leq 0) =$$

$$Y \sim \mathcal{N}(-a, N_0), P(Y > -a/2) =$$

Power Spectrum of Line Codes

$$S_y(f) = |P(f)|^2 \left(\frac{1}{T_b} \sum_n R_x[n] e^{-j2\pi f n T_b} \right)$$

$$R_x[n]$$

Polar Signaling

$$R_x[n] = \delta[n]$$

On-Off Keying (OOK)

$$R_x[n] = \frac{1}{4}(1 + \delta[n])$$

Split-phase, or Manchester coding

$$R_x[n] = \delta[n]$$

Bipolar signaling

$$R_x[n] = \frac{1}{4}(2\delta[n] + \delta[n - 1])$$

*Equal Probabilities

Power Spectrum of Line Codes

$$S_y(f) = |P(f)|^2 \left(\frac{1}{T_b} \sum_n R_x[n] e^{-j2\pi f n T_b} \right) \quad P(f)$$

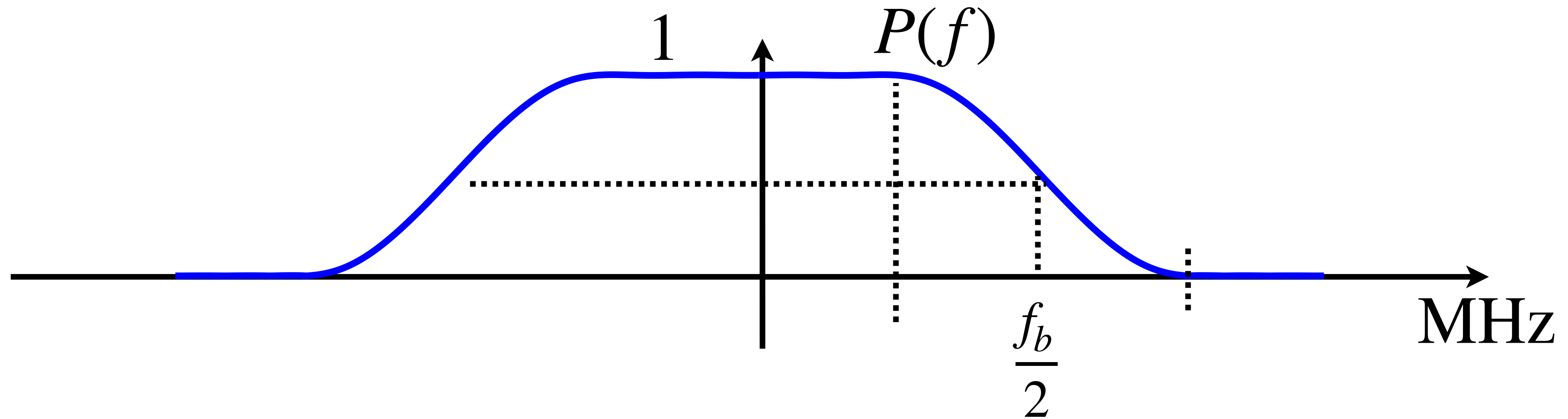
Raised Cosine Pulse

$$P(f) = \begin{cases} T_b, & |f| \leq \frac{1-\beta}{2T_b} \\ \frac{T_b}{2} \left[1 + \cos \left(\left(|f| - \frac{1-\beta}{2T_b} \right) \frac{\pi T}{\beta} \right) \right], & \frac{1-\beta}{2T_b} \leq |f| \leq \frac{1+\beta}{2T_b} \\ 0, & |f| > \frac{1+\beta}{2T_b} \end{cases}$$

Sinc Pulse

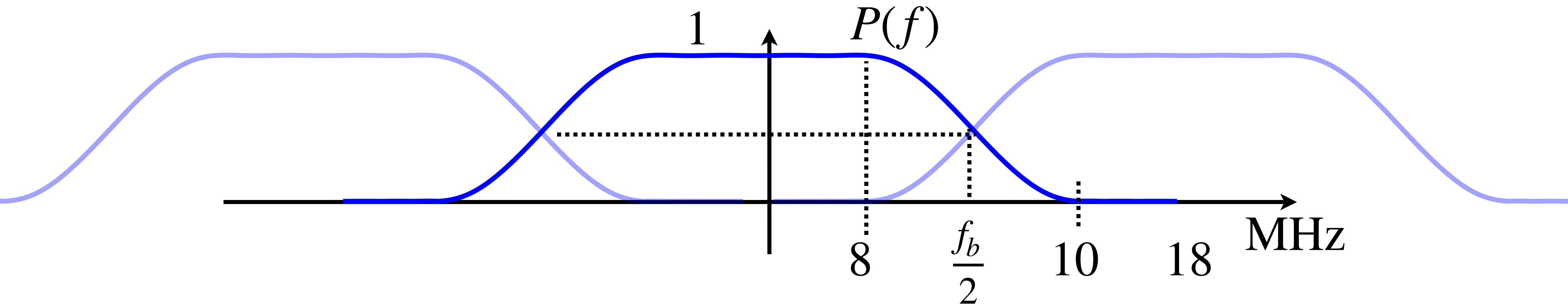
$$P(f) = T_b \Pi(fT_b)$$

Raised Cosine Pulse Practice



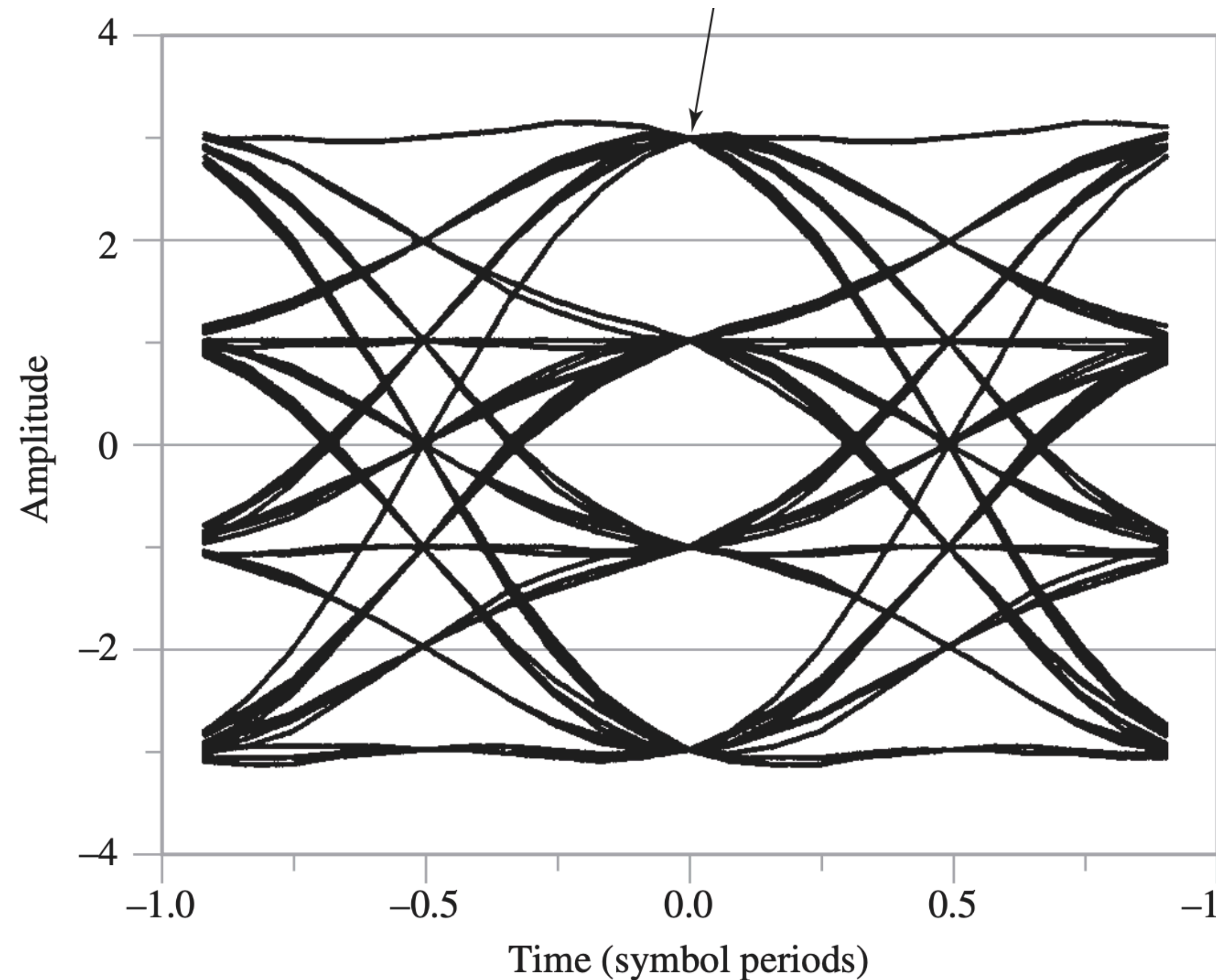
What is the BW using a RC pulse operating at 500Mbps using 64QAM and $\beta = 0.5$?

Raised Cosine Pulse Practice



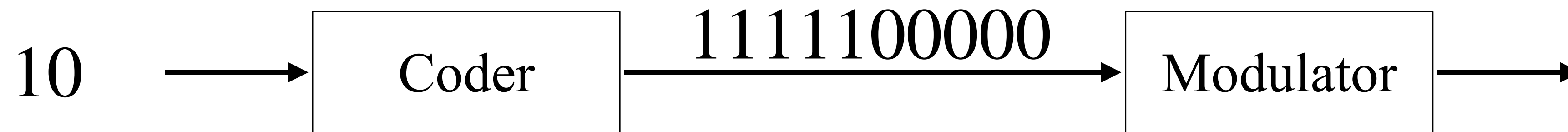
Does this pulse has zero ISI at bitrate of 30Mbps using 8PSK?

Decipher this Eye Diagram



What's the modulation scheme? When should I decide which symbol was sent?
Decision boundaries when each bit is equally likely?

Repetition Codes



What is the code rate?

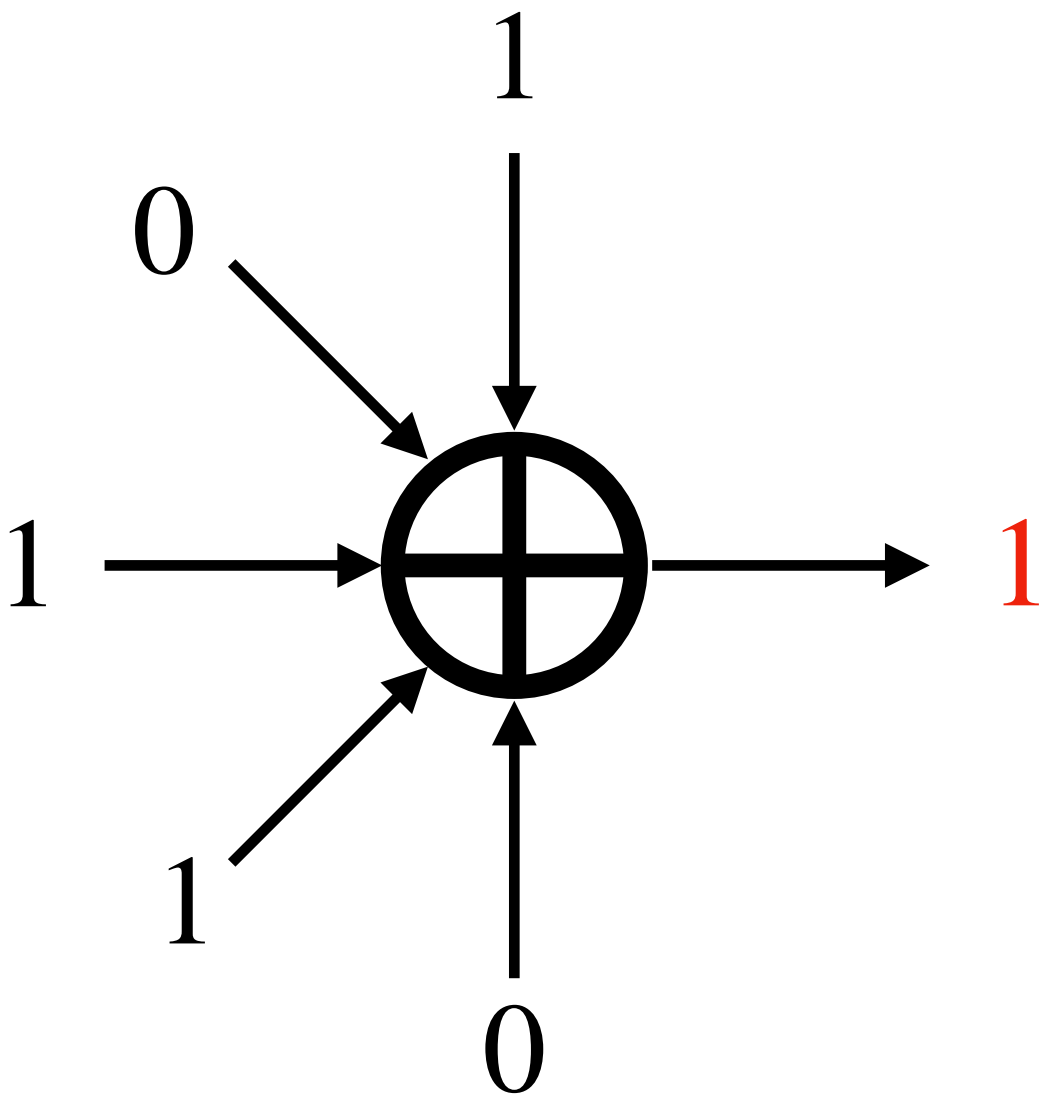
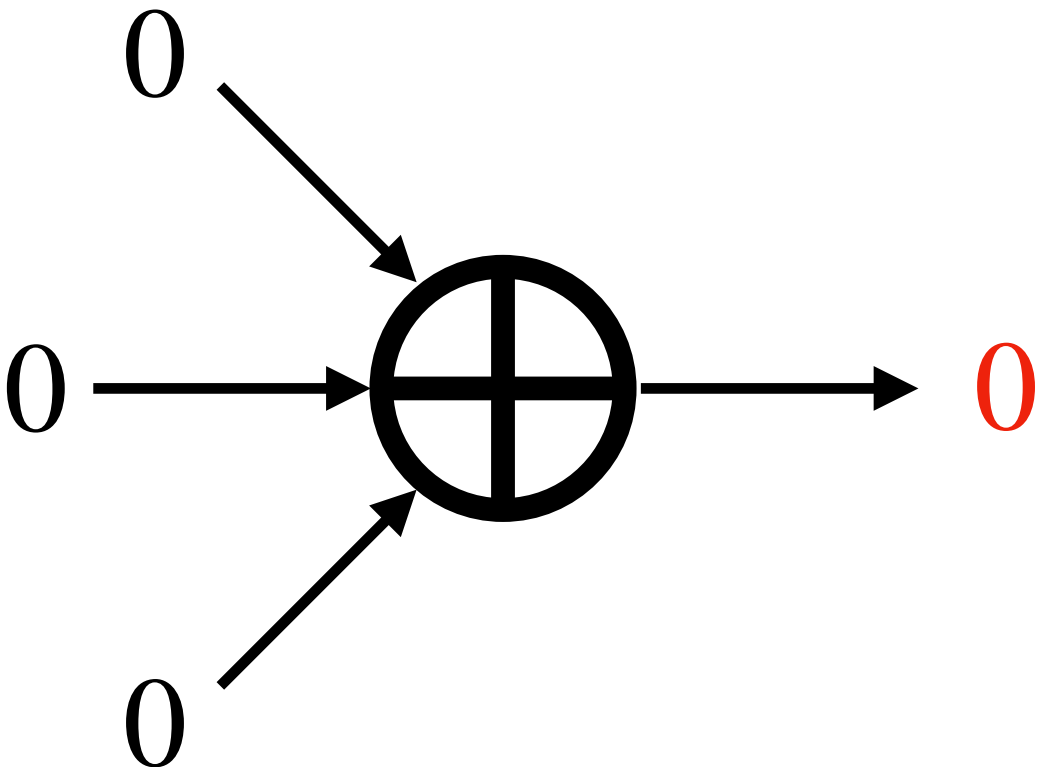
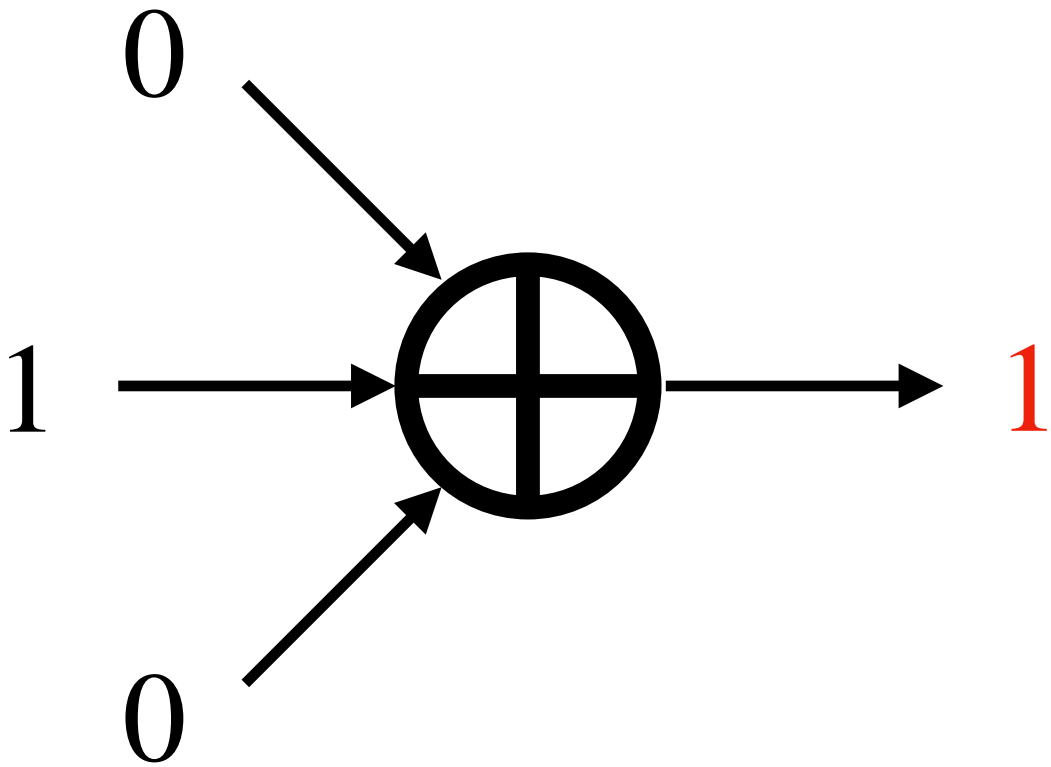
Given that the modulation is PAM2 with $\text{SNR} = 4$, what's the probability of error using a majority vote?

XOR Practice

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

A	B	C	$A \oplus B \oplus C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

ODD number of 1's



Convolutional Codes

Consider a code with generate $G(D)$ given as

$$G(D) = \begin{bmatrix} D & D & 1 + D \\ D & 1 & 1 \end{bmatrix}$$

What's is the code rate?

Draw the corresponding encoder.

Convolutional Codes

Draw the corresponding encoder.

$$G(D) = \begin{bmatrix} D & D & 1 + D \\ D & 1 & 1 \end{bmatrix}$$

Convolutional Codes

Draw the corresponding encoder.